CHAPTER XVI

ON THE EXISTENCE OF RELATIONS

> We cannot choose to do without them, without seeking to choose, since choice is action, and involves, for instance, the aforesaid difference between affirming and denying that we mean to do thus and thus. (449)
>
> JOSIAH ROYCE

In concluding the foregoing remarks, I must explain one more general consideration. This concerns an extremely profound structural psycho-logical discovery, made by Prof. Royce,[1] which underlies any and all semantic problems of human 'mentality'. Royce, although a 'philosopher', was a lover of mathematics and was much interested in the problems of *order*. He was trying to reformulate 'logic' in terms of order. We had already encountered the inherent circularity in the structure of human knowledge, which admittedly is semantically disconcerting if not faced boldly. But, when recognized, this circularity is not only not vicious, but even adds to the interest and beauty of life and makes science more interesting. Besides, the structure of human knowledge is such that there are activities of man which are not only circular but also 'absolute', or 'necessary'. Whatever we do, we cannot get away from them—a fact of serious semantic importance. Except from Royce and a few of his students, these problems have as yet received little attention.

Royce shows that there are certain activities which we reinstate and verify through the very fact of attempting to assume that these forms of activity do not exist, or that these laws are not valid. If any one attempts to say that there are no classes whatsoever in his world, he thereby inevitably classifies. If any one denies the existence of relations, and, in particular, a semantic relation between affirmation and denial, or affirms that 'yes' and 'no' have *one* meaning, in that breath he affirms and denies. He makes a difference between 'yes' and 'no', and emphatically asserts relational equivalence even in denying the difference between 'yes' and 'no'. To use Royce's own remarkable words: 'In brief, whatever actions are such, whatever types of action are such, whatever results of activity, whatever conceptual constructions are such, that the very act of getting rid of them, or of thinking them away, logically implies their presence, are known to us indeed both empirically and pragmatically (since we note their presence and learn of them through action); but they are also absolute. And any account which succeeds in telling what they are has absolute truth. Such truth is a "construction"

or "creation", for activity determines its nature. It is "found", for we observe it when we act.'

We see that we have definite semantic guides in this enquiry. One guide to follow is these unescapable characteristics of the structure of human knowledge, which Royce called 'absolute', but which I prefer to call 'necessary'. The other guide leads us to avoid 'impossible' or absurd statements, or statements which have no 'logical existence'; which, in the rough, means statements which abuse symbolism and produce noises., instead of symbols. As we have already seen, both guides have sound *neurological* justification, to be expressed in terms of *order* and *circularity,* terms uniquely fit structurally to speak about processes, stages of processes, orders of abstractions,. Obviously, our task of formulating a theory of sanity can proceed along these structural and semantic lines. It should be noticed that mathematics, considered as a form of human behaviour, and 'mental' illnesses, also considered as definitely human behaviour, have yielded their share for our structural guidance.

Although many a scientist has instinctively proceeded in the way indicated, yet the instinctive successful procedure of an isolated scientist is usually not capable of being transmitted to others. It is his personal benefit. Only a *methodological structural* formulation of such private routes to semantic success can become a *public* fact, to be analysed, criticized, improved, and transmitted or rejected.

It must be noticed that terms like 'chance' or 'law' are fundamentally connected with discussions of determinism versus indeterminism, and so involve problems in connection with 'necessary' semantic processes. In the example about the probability of the M-event, it was shown how a 'chance' event on one level may become a 'law' on another. The structural *possibility* of such transformations is very interesting and of basic semantic importance. For scientific purposes, we must accept ∞-valued determinism on the scientific level as *it is the test of structure;* but this has nothing to do with the *apparent,* mostly two-valued indeterminism in our daily lives. To solve a number of equations, we must have as many equations as we have unknowns. If we have fewer equations than unknowns, we do not get definite values; our unknowns are still undetermined. The origin of 'indeterminism' is similar; we lack knowledge; the number of equations is less than the number of unknowns. Hence, it is impossible to discover determined values in all cases. This gives an appearance of two-valued indeterminism, but with the increase of our knowledge, or with additional equations, the unknown may be determined. Determinism is a more fundamental point of view than indeterminism; in it we find a *test for structure.* It is also a more general point

of view, in which indeterminism is only a particular case and does not allow of the structural test. In a science of man, in a $\bar{A}$-system, we must start with the more fundamental and general. Accordingly, we have to accept ∞-valued determinism, which, in 1933, becomes the broad scientific point of view. The unnecessary semantic war between the advocates of the different points of view has been unduly bitter and necessarily futile.

As words *are not* the things we speak about, and the only link is structural, the 'human mind' must require linguistic structural ∞-valued determinism as a condition of rationality. As soon as we find that any linguistic issues are not deterministic, it is an unmistakable sign that the language or the 'logic' we are using is *not* similar in structure to the empirical world and so should be changed.

This statement seems to be general. In application to the new quantum mechanics' special problem, it would appear that the old macroscopic language of 'space', 'time'., is not similar to the sub-microscopic structure and should, therefore, be changed. Perhaps the electrodynamic language, instead of the macro-mechanistic, would fare better.

CHAPTER XVII

ON THE NOTIONS OF 'MATTER', 'SPACE', 'TIME'

Common sense starts with the notion that there is matter where we can get sensations of touch, but not elsewhere. Then it gets puzzled by wind, breath, clouds, etc., whence it is led to the conception of "spirit"—I speak etymologically. After "spirit" has been replaced by "gas," there is a further stage, that of the aether. (457) BERTRAND RUSSELL

The supposition of common sense and naive realism, that we see the actual physical object, is very hard to reconcile with the scientific view that our perception occurs somewhat later than the emission of light by the object; and this difficulty is not overcome by the fact that the time involved, like the notorious baby, is a very little one. (457) BERTRAND RUSSELL

We have certain preconceived ideas about location in space which have come down to us from ape-like ancestors. (149) A. S. EDDINGTON

But it does not seem a profitable procedure to make odd noises on the off-chance that posterity will find a significance to attribute to them. (149) A. S. EDDINGTON

There is a blessed phrase "hidden reserves"; and generally speaking the more respectable the company the more widely does its balance-sheet deviate from reality. This is called sound finance. . . .
Thanks to Minkowski a way of keeping accounts has been found which exhibits realities (absolute things) *and balances.* (149) A. S. EDDINGTON

The quest of the absolute leads into the four-dimensional world. (149) A. S. EDDINGTON

The views of space and time which I wish to lay before you have sprung from the soil of experimental physics, and therein lies their strength. They are radical. Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality. (352) H. MINKOWSKI

It is a *thing;* not like space, which is a mere negation; nor like time, which is—Heaven knows what! (149) A. S. EDDINGTON

Newton objectivises space. Since he classes his absolute space together with real things, for him rotation relative to an absolute space is also something real. (151) A. EINSTEIN

Space is only a word that we have believed a thing. (417) H. POINCARÉ

In fact, our ordinary description of nature, and the idea of exact laws, rests on the assumption that it is possible to observe the phenomena without appreciably influencing them. (215) W. HEISENBERG

Even when this arbitrariness is taken into account the concept "observation" belongs, strictly speaking, to the class of ideas borrowed from the experiences of everyday life. It can only be carried over to atomic phenomena when due regard is paid to the limitations placed on all space-time descriptions by the uncertainty principle. (215) W. HEISENBERG

Section A. Structural considerations.

The facts at hand in 1933 show that the language we use for the purpose of describing events *is not* the events; the representation symbolizes what is going on inside our skins; the events are outside our skins and *structural similarity* is the only link between them. Historically, as a race, we learned sooner and more about the events outside our skins than about the events inside our skins; just as a fish or a dog 'knows' a lot about his world, lives sometimes happily and abundantly, and yet 'knows' nothing about biology, or physiology, or psycho-logics. Only recently did we begin to study ourselves scientifically. At some stage of our development, we introduced structurally simple forms of representation, such as a *language* of subject-predicate, of additivity,. We are still perplexed when we find that the events outside our skins cannot be pressed into schemes which are manufactured inside our skins. Our nervous system, with its ordered and cyclic structure and function, manufactures abstractions of different orders, which have quite distinct structure and different characteristics. On different levels, we manufacture different abstractions, dynamic and static, continuous and discontinuous., which have to take care of our needs. If the verbal schemes we invent do not fit structurally the world around us, we can always invent new schemes of new structure which will be more satisfactory. It is not a problem of the world around us, for our words cannot change that, but of *our ingenuity*. In the meantime, we learn something very important; namely, about the world's *structure*, which is the only content of knowledge.

There are good structural reasons why the world should, or should not, be accounted for in terms of differential equations, or in *terms and language* of 'causality',. The term *order* is structurally fundamental and will help us in a radical and constructive way, in our quest.

First, however, we will investigate some further semantic problems, remembering that a theory of sanity, which means a theory of adjustment, should emphasize the methodological and structural means for such semantic adjustment. The dynamic-static translations are fundamentally connected with different orders of abstractions and involve psycho-logical issues connected with 'emotions' and 'intellect', linearity versus non-linearity, 'straight' versus 'curved'., explained in Parts VII and VIII.

In life, as well as in science, we deal with different happenings, objects, and larger or smaller bits of materials. We have a habit of speaking about them in terms of 'matter'. Through a *semantic disturb-*

ance, called identification, we fancy that such a thing as 'matter' has separate physical existence. It would probably be a shock to be invited seriously to *give* a piece of 'matter' (give and *not* burst into speech). I have had the most amusing experiences in this field. Most people, scientists included, hand over a pencil or something of this sort. But did they actually give 'matter'? What they gave *is not* to be *symbolized* simply 'matter'. The object, 'pencil', which they *handed,* requires linguistically 'space'; otherwise, there would be no pencil but a mathematical point, a fiction. It also requires verbally 'time'; otherwise, there would be no pencil but a 'flash'.

Similarly, if any one is invited to *give* a piece of 'space' (again *give* it, and *not* burst into speech), the best he could do would be to wave his hand and try to show 'space'. But the waving of the hand referred to what we call air, dust, microbes, gravitational and electromagnetic fields,. In other words, structurally, the supposed 'space' was *fulness* of some materials already 'in space' and 'in time'.

In the case of *giving* 'time', one could *show* his watch. A similar objection holds, also; namely, that he has shown us so-called 'matter' which is 'moving' in 'space'. It is very important to acquire the *s.r* that when we use the term 'matter' we refer to something, let us say, the pencil, which, according to the accepted *el language,* also involves 'space' and 'time', which we disregard. When we use the term 'space', we refer to a fulness of some materials, which exists in 'time'. But because these materials are usually invisible to the 'senses', we again disregard them. In using the term 'time', we refer to 'matter' moving in 'space', which again we disregard.

What is said here and what will follow is structurally unconditionally fundamental for a theory of sanity, because in most cases of 'insanity' and un-sanity, there is a disorientation as to 'space' and 'time'. In identification, the semantic disturbance which affects nearly all of us, and is at the foundation of the majority of human difficulties, private or public, there invariably appears a special semantic disorientation in our feelings toward 'matter', 'space', and 'time'. This is only natural, for the 'insane' and un-sane are the unadjusted; the 'sane' are the supposedly adjusted.

Adjusted to what? To the world around us and ourselves. Our *human* world differs from the world of animals. It is more complex and the problems of human adjustment become also more subtle. In animal life, attitudes toward the world do not matter in a similar sense; with us, they become important; hence the need of analysis of the new human 'semantic universe', which involves the 'universe of discourse'. This

'universe of discourse' is strictly connected with the *terms* of 'matter', 'space', and 'time', structure, and our semantic *attitude* toward these terms.

Let us return to the analysis of our object which we call the 'pencil'. We have seen that the *object* pencil *is not* 'matter', nor 'space', nor 'time'. A question arises, which has been asked very often and has *never,* to my knowledge, been answered satisfactorily: what 'is' the *object* pencil, and what 'are' the *terms* 'matter', 'space', and 'time'? Here and there some one has given fragments of answers or some satisfactory detached statements. But in every case I know, the semantic disturbance called identification appears, and so even the casual correct answer is not applied but remains enmeshed in some other identifications. I have spent much 'time' and labour in overcoming my own identifications, and now confront the situation that nearly every work I read from this point of view cannot be criticized, but requires rewriting. This task is impossible for me, technically and otherwise. So, finally, I decided to formulate the present $\bar{A}$-system, and then see what kind of reconstruction can be accomplished with the new evaluation.

The answer to the questions set above is childishly simple, yet I will carry it all through and let the semantic consequences speak for themselves. The chunk of nature, the specially shaped accumulation of materials., which we call a pencil, 'is' fundamentally and *absolutely un-speakable,* simply because whatever we may *say* about it, *is not it.* We may write with this something, but we cannot write with its name or the *descriptions* of this something. So the object *is not words.* It is important that the reader be entirely convinced at this point, and it requires some training, performed repeatedly, before we get our *s.r* adjusted to this simple fact. Our statement had two parts. One was rather unpromising; namely, that the object was absolutely un-speakable, because no amount of words will make the object. The other was more promising, for we learned an extremely important, perhaps crucial, semantic fact; namely, what the *object* pencil *is not;* namely, that the *object is not words.* We must face here an important semantic fact. If we are told that we cannot get the moon, we stop worrying about it, and we regard any dream about getting the moon as an infantile phantasy. In this example, we could not even say that such news as the impossibility of getting the moon was sad, or unpleasant, news. We might say so, jokingly, to an infant, but the majority of the grown-ups would not have their *s.r* perturbed by it. A similar situation arises with the object called pencil. The *object* is *not words.* There is nothing sad or depressing about this fact. We accept it as a fact and stop worrying about it, as

an infant would. The majority of the old 'philosophical' speculations about this subject belong to the semantic period of our infancy, when we live in phantasies and structurally gamble on words to which we affectively ascribe objective existence,. This represents full-fledged un-sanity due to identification. The answer to the question, what 'are' the *terms* 'matter', 'space', and 'time', is, as usual, given in the properly formulated question. They 'are' *terms,—'Modi considerandi',* as Leibnitz called them without fully realizing the semantic importance of his own statement. Incidentally, it must be noticed that it was the psycho-logical characteristic of Leibnitz who was capable of such a statement, that was probably responsible for his whole work, as will become more apparent later on. When we abandon primitive standards of evaluation, geniuses will be *made* by a semantic education which relieves the race from the older blockages.

So we see clearly that outside of our skins there is something going on, which *we call* the world, or a pencil, or anything, which is *independent* of our words and which is *not* words. Here we come across a very fundamental irreversible process. We can say that in this world a man and his words have happened. There is a 'causal' eventual complex series between the world, us, and our words, but in unaided nature this process is, in the main, irreversible, a fact unknown to primitives who believe in the magic of words. Through our ingenuity we can make this process partially reversible; namely, we can produce gramophones, telephones in all their developments, electromechanical men who obey orders,.

We know in 1933 that in the semantic world this process is dramatically effective. Words are the result of the activity of one organism, and they, in turn, activate other organisms. On the macroscopic level of ordinary behaviour, this last was known long ago, but only in the last few years has psychiatry discovered what kind of semantic and psychophysiological disasters words and their consequences may produce in the human organism. These last are already on sub-microscopic levels, not obvious and, therefore, only recently discovered.

The structure of the language of 'matter', 'space', and 'time' is ancient. The primitive saw something, ate something, was hurt by something,. Here was an occasion for a grunt of satisfaction, or of pain. The equivalents of words like 'matter', 'substance'., originated. Neither he, nor the majority of us, realized that the small or large bits of materials we deal with appear as extremely complex *processes* (explained in Part X). For him, as for most of us, these bits of materials 'are' 'concrete', whatever that means, and he might know 'all about them', which

must have led to identification., and other delusional evaluations. Of course, these were phantasies of human infancy, and, in lives lived in a world of phantasies, adjustment, and, therefore, sanity, is impossible. Since he did not see or feel, or know about the material he was immersed in, the *fulness* he was living in, he invented the term 'space', or its equivalent, for the *invisible* materials which were present. Knowing nothing of fulness, he objectified what appeared to him to be empty 'space' into 'absolute emptiness', which later became 'absolute space', 'absolute nothingness', by 'definition'.

There are several important remarks which can be made about this 'absolute emptiness' and 'absolute nothingness'. First of all, we now know, theoretically and empirically, that such a thing does not exist. There may be more or less of something, but never an *unlimited* 'perfect vacuum'. In the second place, our nervous make-up, being in accord with experience, is such that 'absolute emptiness' requires 'outside walls'. The question at once arises, is the world 'finite' or 'infinite'? If we *say* 'finite', it *has* to have outside walls, and then the question arises: What is 'behind the walls'? If we say it is 'infinite', the problem of the psychological 'walls' is not eliminated, and we still have the semantic need for walls, and then ask what is beyond the walls. So we see that such a world suspended in some sort of an 'absolute void' represents a *nature against human nature,* and so we had to invent something *supernatural* to account for such assumed nature against human nature. In the third place, and this remark is the most fundamental of all, because a symbol must stand for something to be a symbol at all, *'absolute nothingness' cannot be objective and cannot be symbolized at all.* This ends the argument, as all we may say about it is neither true nor false, but *non-sense.* We can make noises, but say nothing about the external world. It is easy to see that 'absolute nothingness' *is a label for a semantic disturbance,* for verbal objectification, for a pathological state inside our skin, for a fancy, but not a symbol, for a something which has *objective* existence outside our skin.

Some other imaginary consequences of this semantic disturbance are far-fetched and very gloomy. If our world and all other worlds (island universes) were somehow suspended in such an 'absolute void', these universes would radiate their energy into this 'infinite void', whatever that means, and so sooner or later would come to an end, their energy being exhausted. But, fortunately, when we eliminate this pathological semantic state by proper education all these gloomy symptoms vanish as mere fancies. It must be noticed that this 'absolute space', 'absolute void', 'absolute nothingness' with its difficulties, which are due

to very primitive structural speculations on *words,* and to some un-sane ascribing of objectivity to words, can be abolished quite simply if we decide to investigate and re-educate our *s.r.*

What we know positively about 'space' is that it is not 'emptiness', but 'fulness', or a 'plenum'. Now 'fulness' or a 'plenum', first of all, is a term of entirely different *non-el* structure. When we have a plenum or fulness, it must be a plenum of 'something', 'somewhere', at 'some time', and so the *term implies,* at least, *all three of our former elementalistic terms.* Furthermore, fulness by some psycho-logical process does not require 'outside walls'. If we ask if such a universe of fulness is 'finite' or 'infinite', without any psycho-logical difficulties, we may reply that we do not know, but if we study enough of the materials of this universe we *may know.* A universe of fulness may be assumed to have boundaries, and then we may ask again the annoying question: What is beyond? With the proper use of *language,* this difficulty is again eliminated.

Without going into unnecessary details, we see that a boundary, or a limit, or a wall, is something by *definition,* beyond which we cannot go. If there is nothing to restrict our progress, there are *no boundaries.* Let us fancy some cosmic traveller with some extraordinary flying machine, and let us assume that he flies without stopping in a 'definite direction'. If he never encounters any boundary, he is surely entitled to say that his universe is unbounded. The question may arise: Is such an unbounded universe finite or infinite in size? Again let us apply correct language and a little analogy. A traveller on a sphere, like our own earth, could travel *endlessly* without ever coming to a boundary, and yet we know that the sphere, our earth, is of finite size. Mathematicians have worked out this point, and it is embodied in the Einstein theory. The universe is unbounded, an answer which satisfies our feelings; yet it is finite in size, although very large, an answer which satisfies our rationality.* The visualization of such a universe is quite difficult. It should not be visualized as a sphere, but, at a later stage, we will see that it *can* be visualized satisfactorily. The condition for visualization is to eliminate identification, that *semantic disturbance* which is strictly connected with primitive ways of 'thinking'.

The problems of 'time' are similar, although they have a different neurological background. The rough materials we deal with mostly affect our sight, touch, . Invisible materials, like air . , affect these 'senses'

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*I do not introduce here the latest speculations in this field, because, from a non-aristotelian point of view, they appear meaningless.

less, but more the kinesthetic 'senses' by which muscular movements are appreciated, and so 'space' and 'time' have different neurological backgrounds. 'Time' seemingly represents a general characteristic of *all* nervous tissue (and, perhaps, living tissue in general) connected with summarizing or integrating. What we have to deal with in this world and in ourselves appears as periods and periodicity, pulsations,. We are made up of very long chains of atomic pulsating clocks, on the submicroscopic level. On the macroscopic level, we have also to deal with periodic occurrences, of hunger, sleep, breathing, heart-beats,. We know already that, beyond some limits, discontinuous time*s*, when rapid enough, are blended into continuous feelings of pressure, or warmth, or light,. On objective levels we deal with time*s*, and we feel 'time', when the time*s* are rapid enough.

Again, the moving pictures are a good illustration. The normal moving-picture film shows sixteen pictures a second.. The film gives us static pictures with finite differences. When we put it on the projector, the differences vanish. Our nervous system has summarized and integrated them, and we see 'continuous motion'. If pictures are taken at the rate of eight a second and then run on the normal projector for the speed of sixteen a second, we summarize and integrate again, but we see a fast moving picture. If the pictures are taken at the rate of 128 exposures a second and run on the normal projector of sixteen pictures to a second, we have what is called a slow moving picture. It should be noticed that the order of the semantic rhythmic processes is fourfold; it involves order not only in 'space' (three dimensions) but in 'time' also. Periods of contraction alternate with periods of rest, and this occurs at nearly regular intervals.

This rhythmic tendency is, indeed, so fundamental and so inherent in living tissue that we can, at pleasure, make voluntary muscles; for example, exhibit artificially induced rhythmic contractions by immersing them in special saline solutions, as, for instance, a solution of sodium chloride. We should also not wonder why modern science assumes that life may have originated in the sea. The physico-chemical conditions of saline solutions are such that they favour rhythmic processes; they not only may originate them, but may also keep them up, and life seemingly is very closely connected with autonomous rhythmic processes.

Such rhythmic processes are *felt* on lower orders of abstraction as 'continuous time', probably because of the rapidity and overlapping of periods. On higher order abstractions, when structurally proper linguistic and extra-neural means are developed, they appear as time*s*.

Perhaps, neurologically, animals *feel* similarly as we do about 'time', but they have no neurological means to elaborate linguistic and extra-neural means which alone allow us to extend and summarize the manifold experience of many generations (time-binding). They cannot pass from 'time' to 'time*s*'. Obviously, if we do not, we then renounce our human characteristics, and copy animals in our evaluating processes, a practice which must be harmful.

In nature the visible and invisible materials seemingly consist of recurring pulsations of extremely minute and rapid periods, which, in some instances, become macroscopic periods. In the first case, we cannot see them or feel them, so we talk about 'concreteness',. In the second case, we see the periodic movements, as of the earth around the sun., or we feel our heart-beats,. We see that the visible or invisible materials in nature are compounded of periodic pulsations and are simply two aspects of one process. The splitting of these processes into 'matter', 'space', and 'time' is a characteristic function of our nervous system. These abstractions are *inside* our skins, and are methods of representation for ourselves to ourselves, and *are not* the objective world around us.

It must be realized that under such circumstances we cannot speak about 'finiteness' or 'infiniteness' of 'matter', 'space', and 'time', as all the old 'philosophers' have done, Leibnitz included, because these terms 'finite' and 'infinite', though they may be conceivably applied to *numbers* of aspects of objective entities, have *no meaning* if applied to linguistic issues, that is, to *forms of representation* outside of numbers. Of course, if, through a semantic pathological disturbance (objectification), we do ascribe some delusional objective existence to verbal terms, we can then talk about anything, but such conversations have no more value than the deliria of the 'mentally' ill. The terms 'finite' or 'infinite' are only legitimately applied to *numerical* problems, and so we can speak legitimately of a finite or infinite numbers of inches, or pounds, or hours, or similar entities, but statements about the 'finite mind' or the 'understanding of the infinite'., have *no* meanings and only reveal the pathological semantic disturbance of the patient.

The objectification of *our feeling* of 'time' has had, and has at present, very tragic consequences strictly connected with our un-sanity. It must be remembered that particularly in 'mental' and nervous difficulties the patient seldom realizes the character of his illness. He may feel pains, he may feel very unhappy, and what not, but he usually does not understand their origin. This is particularly true with semantic disturbances. One may explain endlessly, but, in most cases, it is perfectly hopeless to try to help. Only a very few benefit. Here lies, also, the

main difficulty in writing this book. Readers who identify, that is, who believe unconsciously with all their affective impulses in the objectivity of 'matter', 'space', and 'time', will have difficulty in modifying their *s.r* in this field.

Let us see now what consequences the objectification of 'time' will have for us. If we do *not* objectify, and *feel* instinctively and permanently that words *are not* the things spoken about, then we could not speak about such *meaningless* subjects as the 'beginning' or the 'end' of 'time'. But, if we are semantically disturbed and objectify, then, of course, since objects have a beginning and an end, so also would 'time' have a 'beginning' and an 'end'. In such pathological fancies the universe must have a 'beginning in time' and so must have been made., and all of our old anthropomorphic and objectified mythologies follow, including the older theories of entropy in physics. But, if 'time' is only a *human form of representation* and *not an object,* the universe has no 'beginning in time' and no 'end in time'; in other words, the universe is 'time'-less. It was not made, it just 'was, is, and will be'. The moment we realize, feel permanently, and utilize these realizations and feelings that words *are not* things, then only do we acquire the semantic freedom to use different forms of representation. We can fit better their structure to the facts at hand, become better adjusted to these facts which *are not* words, and so evaluate properly *m.o* realities, which evaluation is important for sanity.

According to what we know in 1933, the universe is 'time'-less; in other words, there is no such *object* as 'time'. In terms of periods, or years, or minutes, or seconds, which is a *different language,* we may have infinite numbers of such time*s*. This statement is another form of stating the principle of conservation of energy, or whatever other fundamental higher abstraction physicists will discover.

Because 'time' is a *feeling,* produced by conditions of this world outside and inside our skins, which can be said to represent time*s*, the problem of 'time' becomes a neuro-mathematical issue. It must also be noticed that time*s*, as a term, implies time*s* of something, somewhere, and so, as with plenum or fulness, it is structurally a *non-el*, $\bar{A}$ term.

Time*s* has also many other most important implications. It implies *numbers* of time*s*, it implies periods, waves, vibrations, frequencies, units, quanta, discontinuities, and, indeed, the whole structural apparatus of modern science.

Euclidean 'space' had the semantic background of 'emptiness'. In it we moved our figures from place to place and always assumed that this could be done quite safely and accurately. Newtonian mechanics

also followed this path and even postulated an 'absolute space' (emptiness). All of which harks back to the old aristotelianism.

$\bar{E}$, $\bar{N}$, and $\bar{A}$ systems have the semantic background of fulness or plenum, although, unfortunately, this background is, as yet, mainly unrealized, not fully utilized; it has not, as yet, generally affected our *s.r.*

A simple illustration will make the difference clear. Imagine that in one part of a large room we have an open umbrella which we would like to compare with another 'unit' open umbrella. Let us imagine that the room has the air pumped out and also that all other eventual disturbing factors are eliminated. We can move our open 'unit' umbrella from one part of the room to another, and this movement will not considerably distort our 'unit' umbrella. Now let us perform a similar experiment in two houses, separated some distance, during a storm, a storm implying, of course, *fulness*. Can we transport our 'unit' umbrella through the storm and preserve its shape., in a fulness, without taking the fulness into account? Of course not. We see what serious difference it makes if our theories presuppose 'emptiness' or 'fulness'.

This shows also why the non-euclidean geometries which deal with a plenum are structurally preferable and semantically sounder and more in accord with the structure of the world, than the language of euclidean 'emptiness', to which there is nothing in nature to correspond. Should we wonder that modern linguists (mathematicians) work in the direction of fulness and of fusing geometry with physics. It is obviously the only thing to do. Differential geometry is the foundation of this new outlook, but, even in this geometry, lines could legitimately be transported over great distances. Weyl introduced a semantic improvement of this point of view by assuming that for a differential geometry it is illegitimate to use comparisons at large distances, but that all operations should be between indefinitely near points.[1]

It should be noticed that scientists, in general, disregard almost completely the verbal and semantic problems explained here, a fact which leads to great and unnecessary confusion, and makes modern works inaccessible to the layman. Take, for instance, the case of the 'curvature of space-time'. Mathematicians use this expression very often and, inside their skins, they know mostly what they are talking about. Millions upon millions of even intelligent readers hear such an expression as the 'curvature of space-time'. Owing to nursery mythology and primitive *s.r*, 'space' for them is 'emptiness', and so they try to understand the 'curvature of emptiness'. After severe pains, they come to a very *true*, yet, for them, hopeless, conclusion; namely, that 'curvature of emptiness' is either *non-sense* or 'beyond them', with the semantic result that either

they have contempt for the mathematicians who deal with non-sense or feel hopeless about their own capacities—both undesirable semantic results.

The truth is that 'curvature of emptiness' has no meanings, no matter *who* might say it, but curvature of fulness is entirely different. Let the reader look at the cloud of smoke from his cigarette or cigar, and he will at once understand what 'curvature of fulness' means. Of course, he will realize, as well as the mathematicians do, that the problem may be difficult, but, at least, it *has sense* and represents a problem. It is not non-sense.

Similar remarks apply to higher dimensions in 'space'. Higher dimensions in 'emptiness' is also non-sense; and the layman is right in refusing to accept it. But higher dimensions in fulness is entirely a different problem. A look at the cloud of smoke from our cigarette will again make it completely plain to everybody that to give an account of fulness, we may need an enormous number of data or, as we say roughly, of dimensions. This applies, also, to the new four-dimensional world of Minkowski. It is a fulness made up of world lines, a network of events or intervals., and it is not non-sense.

Lately, there has appeared an excellent book by Bertrand Russell, published by the International Library of Psychology, Philosophy and Scientific Method; and yet the title is *The Analysis of Matter,* without any quotation marks.

This book is really an unusually fine and fundamental work which has no defects which could be implied by the title. This title simply disregards the issues explained here; it should be *The Analysis of 'Matter'*.

It is with some pleasure that one sees such an authority as Eddington, in his *The Mathematical Theory of Relativity,* on p. 158, making the statement: 'In using the word "space" it is difficult to repress irrelevant ideas; therefore let us abandon the word and state explicitly that we are considering a *network of intervals'*.

For the reasons already given, I do not use the terms 'matter', 'space', or 'time' without quotation marks; and, wherever possible, shall use, instead, the terms 'materials', 'plenum', 'fulness', 'spread', and 'time*s*', (say seconds). Indeed, these semantic problems are so serious that they should be brought to the attention of International Mathematical and Physical Congresses, so that a new and *structurally correct* terminology could be established. It is *not* desirable that science should *structurally mislead* the layman and disturb his *s.r.* It is easier for trained specialists to change their terminology than to re-educate semantically the rest of the race. I would suggest that terms 'matter', 'sub-

stance', 'space', and 'time' should be completely eliminated from science, because of their extremely wide-spread and vicious structural and so semantic implications, and that the terms 'events', 'space-time', 'material', 'plenum', 'fulness', 'spreads', 'times'. , be used instead. These terms not only do not have the old structural and semantic implications, but, on the contrary, they convey the *modern* structural notions and involve new *s.r.* The use of the old terms drags in, unconsciously and automatically, the old primitive metaphysical structure and *s.r* which are entirely contradicted by experience and modern science. I venture to suggest that such a change in terminology would do more to render the newer works intelligible than scores of volumes of explanations using the old terminology.

Before summarizing in Parts IX and X what modern science has to tell about the structure of the world around us, it will be profitable to enquire what are the means by which we can recognize this structure.

Section B. The neurological function of abstracting.

Protoplasm, even in its simplest form, is sensitive to different mechanical and chemical stimulations; and, indeed, undifferentiated protoplasm has already all the potentialities of the future nervous system. If we take an undifferentiated bit of protoplasm, and some stimulation is applied to some point, the stimulus does not spread somehow 'all over at once', with some mysterious 'infinite' velocity, but propagates itself with finite velocity and a diminishing gradient from one end of our bit of protoplasm to the opposite end.

Because of the *finite velocity* of propagation and the fact that the *action is by contact in a plenum,* the impulse has a definite direction and diminishing intensity, or, as we say, the bit of protoplasm acquires a temporal polarity (head-end). Such polarity conditions produce a directed wave of excitement of diminishing intensity, which Child calls a dynamic gradient. If such a stimulation were applied to one spot for a considerable length of 'time', some kind of polarization may become lasting. In some such way those dynamic gradients have become *structuralized* in the forms of our nervous system, which represent the preferred paths by which the nervous impulses travel.

The bodies of most organisms are protected from outside stimulation by some kind of membrane or cuticle and the parts of the surface have developed so as to be sensitive to one form of stimulation and not to others. For instance, the eye registers the stimulations of light waves, while it is insensitive to sound . , and, even if hit, it gives only the feeling of light. Each 'sense-organ' has also the nervous means of concentrating

stimuli, intensifying them . , and so of effecting the most efficient response of the corresponding end-organ.

In our school days we were taught that we have five 'senses'. Modern researches show that there are more than twenty different 'senses'. Besides, as far as 'Smith' is concerned, we know that 'senses' and 'mind' cannot be divided.

The main stimulations which we find in the outside world may be divided into three groups. The first are connected with the roughest macroscopic manifestations of the outside world; they are mechanical impacts which we abstract as 'tactile sensations', which range from a single mechanical contact to rhythmically repeated contacts with our skin as frequently as 1552 vibrations per second. Above this limit 'times' begin to be registered as a 'duration'; that is, the individuality of 'times' is lost, and we feel pressure. At this level we deal with gross macro scopic manifestations, which are not only felt but can also be seen.

The second group of manifestations is, in the main, no more on the gross level. Here belong the vibratory manifestations which are no more visible to the unaided eye. We may speak of them as on the microscopic level. They are mechanical vibrations of the air . , and we become acquainted with them in the form of sound. The vibrations which the average ear is able to register range from about 30 (sometimes even 12) to about 30,000 or even 50,000 vibrations a second. The ear does not register any other vibrations.*

The third group of vibratory manifestations belongs to a still subtler level. They are electromagnetic waves of an enormous variety of wave-lengths and number of vibrations per second. The lower members of this series are the Hertzian electric waves, the higher members are the X-, or Röntgen-rays. Our nervous systems are capable of registering only a very limited range of these vibrations; namely, the waves called radiant heat, the light waves, and the ultra-violet rays, these last only on a chemical level. It seems that we have no organ which responds directly to electric waves, ultra-violet rays, X-rays, and the many other rays which we know from laboratory work.

Similarly, the chemical 'senses' of taste and smell register only a very small number of actual excitations to which they are exposed.

Animals have different limits of nervous susceptibility, but we can have no idea how the world looks to them unless their nervous system is quite similar to our own. The above statements will become clearer if we tabulate some of them. The following table is taken from *An Introduction to Neurology* by Professor C. Judson Herrick, p. 85 (Fifth Edition):

*Latest researches seem to modify these data.

Table of Physical Vibrations*

Physical process.	Wave length.	Number of vibrations per second.	Receptor.	Sensation.
Mechanical contact.	. . . . .	From very slow to 1552 per second.	Skin.	Touch and pressure.
Waves in material media.	Above 12,280 mm.	Below 30 per second.	None.	None.
	12,280 mm. to 13 mm.	30 per second to 30,000 per second.	Internal ear.	Tone.
	Below 13 mm.	Above 30,000 per second.	None.	None.
Ether waves.	∞ to .2 mm. (electric waves).	0 to 1500 billion (1.5×10^{12}).	None.	None.
	.1 mm. to .0004 mm.	3000 billion (3×10^{12}) to 800,000 billion (8×10^{14}).	Skin.	Radiant heat.
	.0008 mm. to .0004 mm.	400,000 billion (4×10^{14}) to 800,000 billion (8×10^{14}).	Retina.	Light and color.
	.0004 mm. to .000008 mm. (ultra-violet rays).	800,000 billion (8×10^{14}) to 40,000,000 billion (4×10^{16}).	None.	None.
	.00002 mm. to .00000001 mm. (x-rays).	15,000,000 billion (1.5×10^{16}) to 30,000,000,000 billion (30×10^{18}).	None.	None.
	.00000014 mm. to .0000000005 mm. (γ-rays).	2 billion billion (2×10^{18}) to 600 billion billion (6×10^{20}).	None.	None.
	.00000000005 mm. to .000000000008 mm. (cosmic rays).	6,000 billion billion (6×10^{21}) to 40,000 billion billion (4×10^{22}).	None.	None.

*The use of names for large numbers is not uniform in different countries, and so I give, in brackets, the United States and French equivalents to the English names.

Million 1,000,000 $= 10^6$; (million).
Milliard 1,000,000,000 $= 10^9$; (billion or milliard).
Billion $10^6 \times 10^6 = 10^{12}$; (trillion).

In this table 1 billion $= 10^9$.

As a further illustration of the mechanism of abstracting, we may suggest the observation of Weber that if, for instance, a room is lighted with 100 candles, and if one more candle is brought in, the increased illumination will be appreciated very slightly. But not so if we had a room illuminated with 1000 candles. In this case, we should not appreciate the addition of one candle at all. Ten candles should be introduced to make an appreciable difference in our perceptions. The Weber law, as it is called, stated that in the above case 1/100 of the original strength of the stimulus is needed to make a change appreciable. For light, the fraction is about 1/100; for noise, about 1/3; for pressure, it varies between 1/30 and 1/10; for weight, between 1/70 and 1/40 in various parts of the body.

If we use compasses and experiment with pricks, we find that in different parts of the body the limit of the distance apart of the points when we *feel one prick* and yet have two, is different.

On the tip of the tongue this limit is	1 mm.
On the palmar surface of third phalanx of forefinger	2 mm.
On the palmar surface of second phalanges of fingers	4 mm.
On the palm of the hand	10 mm.
On the dorsal surface of first phalanges of fingers	14 mm.
On the back of hand	25 mm.
On the upper and lower parts of forearm	37 mm.
On the middle thigh and back	62 mm.[2]

A 'sensation' requires appreciable 'time' (time*s* by a clock) for its development. Part of the 'time' is spent at the end-organ, part in conveying the nervous impulse along the nerves to the brain and part in the brain. A 'sensation' usually outlasts the stimulus, and often a single stimulus produces a whole series of 'after-sensations'.

As compared with the 'sensations' obtained from pain spots, touch is quicker in its development and persistence. With a vibrating string, 1500 vibrations a second are recognizable by touch as vibrations. At over 1552 vibrations a second, the vibratory character is lost, and we feel only continuous pressure. A toothed revolving wheel gives the feeling of smoothness (and 'continuity') when the teeth meet the skin at the rate of from 480 to 680 per second.[3]

The above given tables and facts are deeply significant. We see, first of all, that structurally we are immersed in a world full of energy manifestations, out of which we abstract directly only a very small portion, these abstractions being already coloured by the specific functioning and structure of the nervous system—the abstractors. Very probably, there are many more energy manifestations which, as yet, we have not

discovered. Every few years we discover some new form of energy manifestation, and, at present, our knowledge is already so advanced that it is highly probable that the list is much longer. Finally, and here the whole 'structure of human knowledge' begins to play its role; for sanity *we have to know and evaluate this world* around us, if we want to adjust ourselves satisfactorily to it.

Section C. Problems of adjustment.

Is the problem of adjustment in the animal world similar to that in the human world? No, it is entirely different. Animals do not alter their environment so rapidly, nor to such an extent as humans do. Animals are not time-binders; they have not the capacity by which each generation can start where the former left off. Neurologically, animals have no means for extra-neural extensions, which extensions involve the complex mechanism with which we are dealing throughout this work.

The example of the caterpillar, already cited, shows clearly how organisms not adapted to their environment perish and do not propagate their special, non-survival characteristics. Similar remarks apply to hens, their eggs, and chicks which are kept in buildings without sunlight or with ordinary glass windows; these, also, do not survive, and so pass out of the picture.

With humans, the situation is entirely different. We are able to produce conditions which do not exist in unaided nature. We produce artificial conditions and so *our numbers and distribution* are not regulated by unaided nature alone. Animals cannot over-populate the globe, as they do not produce artificially. We do over-populate this globe because we produce artificially. With *animals, selfishness comes before altruism, and the non-selfish perish. An animal has to live first, then act.* With man, the reverse is true. The selfish may produce such conditions that they are destroyed by them. We can over-populate the globe because of artificial production, and so we are actually born nowadays into a world where we must *act first before we can live.* As I have already shown in my *Manhood of Humanity* (p. 72), the old animalistic, fallacious generalizations have been, and are, the foundation of our 'philosophies', 'ethics', systems., and naturally such animalistic doctrines must be disastrous to us. Neurologically, we build up conditions which our nervous systems cannot stand; and so we break down, and, perhaps, shall not even survive.

Animals have no 'doctrines' in *our meaning* of the term; thus, doctrines are no part of their environment, and, accordingly, animals cannot perish through false doctrines. We do have them, however, and,

since they are the most vital environmental semantic conditions regulating our lives, if they are fallacious, they make our lives unadjusted and so, ultimately, lead to non-survival.

So we see that 'human adjustment' is quite a different and much more complex affair than 'animal adjustment'. The 'world' of 'man' is also a different and much more complex 'world' than that of the animal. There seems to be no escape from this conclusion. We see, also, that what we used to call 'senses' supply us with information about the world that is very limited in quantity, *specific* in quality, an abstraction of low order, never being 'it'. Being often unaffected, our 'senses' are not able to abstract, obviously, some of the most fundamental manifestations of energy to be found in the external world. If we speak of the older so-called 'sense perceptions' as lower order abstractions, then we find that we learn about the other subtler manifestations of energy through science, higher neural and extra-neural means, which we may call higher order abstractions. In the older days, we called this kind of knowledge 'inferential knowledge'. The animals do not have these higher order abstractions in that sense, and so their world is for them devoid of these extra-sensible manifestations of energy.

It must be remembered that these higher order abstractions and the 'inferential knowledge' of the old theories (they are not equivalent by definition) have a very similar status. Organisms work as-a-whole, and to separate completely higher and lower order abstractions is impossible. All that is said here justifies the new terminology. Our nervous system does abstract, does summarize, does integrate on different levels and in different orders, and the *result* of a stimulus *is not* the stimulus itself. The stone *is not* the pain produced by the stone dropping on our foot; neither is the flame we see, nor the burn we feel. The actual process goes on outside of our skin, as represented by the 'realities' of modern science.

We have already spoken frequently of the different order abstractions, their special characteristics, dynamic versus static., and the means of translation of lower orders into higher, and vice versa. Events which are going on and for which we have no direct 'senses' of abstraction, as, for instance, electric waves, Röntgen-rays, wireless waves., we know only through extra-neural extensions of our nervous system given by science and scientific instruments. Naturally, we should expect that the structure of our abstracting mechanism would be also reflected in these higher order abstractions. Facts show this to be true; and practically all modern science proves it directly or indirectly. This is why, for instance, we have the mathematical methods for passing from dynamic

to static, and vice versa; why we have quantum theories and conditions; and why we have problems of continuity versus discontinuity, atomic theories,.

The above is not a plea for certain old-fashioned 'idealistic philosophies' and still less for 'solipsism'. Far from it. The object of this present work is to face hard structural experimental *m.o* facts, analyse these facts in a language of a similar *structure* ($\bar{A}$), and so to reach tentatively new conclusions which again can be verified by experiments. Once more the reader must be warned against carelessly translating the structurally *new* terms into the *old* terms. The complete structural, psycho-logical, semantic, and neurological analysis of one single such new term would afford material for several volumes and so is impossible in this work. The usefulness of the old terms has been exhausted. The structural consequences of the old terms have been practically all worked out, and, as a rule, we cannot have much quarrel with the older conclusions in the *old language*. If we reach *different* conclusions, or get some new emphasis, it will be due to the use of the structurally *new* language. If we translate the new into the old, the old conclusions are usually *truer* than the new ones. The reverse is also true; the old conclusions become false or, at best, only gain emphasis because of the structure of the new language. The problem of all theories, old or new, is to give a structural account of the facts known, to account for exceptions, and to predict new experimental structural facts which again may be verified empirically.

Section D. Semantic considerations.

We speak much and vaguely about the 'structure' of language, but extremely little work has been done in this field. In the present work, we not only tackle this problem as best we can, theoretically, but we also use a language of a new *non-el,* functional structure, and the results, whatever their value, are actually the results of such procedure.

A short while ago we did not even know that such problems existed. Dreams, alone, about such problems did not help, for, before the structures of two different entities can be compared, these entities must first be produced. Then, and only then, can we compare and evaluate them. Before we could compare the A, the E, and the N systems with $\bar{A}$, $\bar{E}$, and $\bar{N}$ systems the last had to be produced, no matter how imperfect they might be at the beginning.

Something similar can be said about languages. Before we can speak of them in the plural and compare them, we must have more than one for comparison. Mathematics has pointed out this problem for us in

geometry. For instance, we have to deal with different frames or references, or different systems of co-ordinates. We find that they represent different languages and that they may introduce purely verbal statements which have nothing to do with the subject of our analysis (extrinsic characteristics). We have also found that some characteristics may appear in one form of representation and not appear so readily in another. For instance, we know that every line, except the X axis through a point 0 which is the intersection of a parabola with its X axis, cuts the curve a second time. This fact, important for us to know, appears clearly in the polar co-ordinate form of representation, but does not appear in the rectangular form of equations, although, when once a characteristic of a curve is discovered, it can be usually translated into the other co-ordinate languages. In such cases as this, a language of new structure has a kind of creative character, in that it makes some structural discoveries easier.

But the co-ordinate methods were not quite satisfactory. They introduced, too easily, too many extraneous, extrinsic characteristics, belonging to the language and not to the subject. Mathematicians decided to get away from these metaphysical 'outside' references by referring the entity to itself to become more experimental. They invented the internal theory of surfaces, a vector language where they *refer the entity to itself,* its curvature, its length and direction. Finally, in the extension of the vector language which is called the tensor calculus, they achieved a still larger kind of independence. Having invented *three* languages in which we can speak about *one* issue, we are now able to meet the problem of *comparison* of these languages. At once, most important structural and methodological problems arise.

The newer quantum mechanics are also an epoch-making linguistic structural innovation, not only in physics, but also in *methodology.* We have, at present, three, or, perhaps, more, quantum mechanics which speak about one subject, but in entirely different languages. I say 'three or more', because, from a methodological point of view, it is very hard, at present, to be precise, as the problems are too new and, as yet, too little analysed. Similar remarks apply to systems. Before two systems can be compared, a second system must be produced. Then we can compare them.

In our brief verbal analysis of 'space', and 'time', and 'matter', we have seen that these represent *terms,* or *linguistic means, not objects.* We have seen, also, that these antique forms of representation have very unsatisfactory structural implications. They introduce a verbal *elementalism* structurally *absent* in nature, and by a process of objecti-

fication lead to many kinds of fanciful semantically harmful metaphysics. Since Einstein and Minkowski, the excellent term 'event' has been introduced into scientific literature. It is a term of such epoch-making semantic importance that it should become a term of daily use and should be introduced into elementary schools. Teachers do not perform their duties honestly or intelligently if they disregard such structural, linguistic, and semantic issues, which, as we have seen, are the central problem of all possible education.

Likewise, we have already seen that the chunk of nature which we call a 'pencil' *is not* 'matter' nor 'space' nor 'time', the terms being only *terms*. Is such *el language* structurally appropriate for the purpose of speaking about the world around us? It seems undeniable that such language is quite out-dated and very unsatisfactory. It introduces structurally an artificial elementalism of a verbal character, in spite of the fact that even the most elementary consideration shows that structurally the opposite is true; namely, that 'matter', 'space', and 'time' can *never* be experimentally divided. The term 'event' is precisely the term which does away with this old and vicious elementalism.

All that we deal with in the outside world involves indivisibly 'matter', 'space', and 'time'. Using the old language, there cannot be something somewhere at 'no time', or something at some 'time' and 'no where' or 'nothing' 'somewhere' at 'some time'. Everything which happens must be structurally represented as something, somewhere, at some 'time'. If the structure of the world happened to be such that 'nothing' would happen 'nowhere' at 'no time', then we should have nothing to talk about, and all we would or could say would deal with our fancies. The four-dimensional language, which describes happenings structurally more nearly as we experience them, is precisely the language of 'events'. It should be remembered that in daily life we live by four-dimensional event-conditions. That is, the events which interest us are something, somewhere, and some 'time'. If we want two of our friends to become acquainted with each other, we invite them to our home. The appointment is in three dimensions in 'space' (to the left or to the right, forward or backward, up or down), and at a given hour. So we see that our daily life is lived in a four-dimensional space-time manifold, and we begin to appreciate the fact that science has lately caught up with such fundamental structural 'realities'. It must be noticed that the new four-dimensional space-time *language* does not, or should not, use the *term* 'matter' as we used it in the old way. In the new language, the bits of materials we deal with are connected analytically with the 'curvature' of this space-time manifold.

The reader should realize that the structurally new language is similar to the structure of our experience, and involves profound methodological and, therefore, psycho-logical, semantic factors. It has entirely different semantic values; and, perhaps, because of this fact, it is an irreversible advance, no matter how details may be altered.

The newtonians, for the most part, overlook the fact that all theories, their own included, are a semantic product of the functioning of the nervous system, and so involve some 'logic' and 'psychology'. In the new theories, a kind of *physical* subjectivity always appears which ought to be taken care of. We know, for instance, that if we immerse a part of a straight stick in water, the stick appears to be broken, although actually it has not been broken. A photographic camera gives a similar record. So we see that, besides the psycho-logical subjectivity, there is a most important *physical subjectivity,* which is introduced by the use of instruments. The main difficulties in modern science are precisely in the elimination of this physical subjectivity, particularly when we deal with such minute entities that the light waves miss them. In the case of an hypothetical gamma-ray microscope, for instance, the rays would produce what is called a Compton effect,* and so the results of the experiment would be altered by the instrument and procedure.

We ought not to be surprised that the old systems of 'motion' and 'emotion' in science (Newton) and 'philosophy' (Bergson) should result from speculations on the old *A el* language and the introduction of fanciful and fallacious assumptions of an 'infinity' somewhere, and other fancies. The realization of this marks a new semantic epoch in our lives. It is to the credit of these two men that they have summarized these old tendencies in such a masterly way that we are enabled to go beyond them. We shall return to this subject when we analyse the four-dimensional 'world' of Minkowski, and, then, we shall summarize briefly what we now know about 'space', 'time', and 'matter' (see Parts IX and X).

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*Compton discovered, in 1923, that the generation of secondary continuous Röntgen radiation by a primary radiation is accomplished by an increase in the wave-length.

PART V

ON THE NON-ARISTOTELIAN LANGUAGE CALLED MATHEMATICS

Once a statement is cast into mathematical form it may be manipulated in accordance with these rules and every configuration of the symbols will represent facts in harmony with and dependent on those contained in the original statement. Now this comes very close to what we conceive the action of the brain structures to be in performing intellectual acts with the symbols of ordinary language. In a sense, therefore, the mathematician has been able to perfect a device through which a part of the labor of logical thought is carried on outside the central nervous system with only that supervision which is requisite to manipulate the symbols in accordance with the rules. (583)

HORATIO B. WILLIAMS

The toughminded suggest that the theory of the infinite elaborated by the great mathematicians of the Nineteenth and Twentieth Centuries, without which mathematical analysis as it is actually used today is impossible, has been committing suicide in an unnecessarily prolonged and complicated manner for the past half century. (22)

E. T. BELL

The solution goes on famously; but just as we have got rid of the other unknowns, behold! V disappears as well, and we are left with the indisputable but irritating conclusion—

$$0 = 0$$

This is a favourite device that mathematical equations resort to, when we propound stupid questions. (149)

A. S. EDDINGTON

Who shall criticize the builders? Certainly not those who have stood idly by without lifting a stone. (23)

E. T. BELL

. . . let me remind any non-mathematicians . . . that when a mathematician lays down the elaborate tools by which he achieves precision in his own domain, he is unprepared and awkward in handling the ordinary tools of language. This is why mathematicians always disappoint the expectation that they will be precise and reasonable and clear-cut in their statements about everyday affairs, and why they are, in fact, more fallible than ordinary mortals. (529)

OSWALD VEBLEN

CHAPTER XVIII

MATHEMATICS AS A LANGUAGE OF A STRUCTURE SIMILAR TO THE STRUCTURE OF THE WORLD

To-day there are not a few physicists who, like Kirchhoff and Mach, regard the task of physical theory as being merely a mathematical description (*as economical as possible*) of the empirical connections between observable quantities, *i. e.* a description which reproduces the connection, as far as possible, without the intervention of unobservable elements. (466) E. SCHRÖDINGER

But in the prevalent discussion of classes, there are illegitimate transitions to the notions of a 'nexus' and of a 'proposition.' The appeal to a class to perform the services of a proper entity is exactly analogous to an appeal to an imaginary terrier to kill a real rat. (578) A. N. WHITEHEAD

Roughly it amounts to this: mathematical analysis as it works today must make use of irrational numbers (such as the square root of two); the sense if any in which such numbers exist is hazy. Their reputed mathematical existence implies the disputed theories of the infinite. The paradoxes remain. Without a satisfactory theory of irrational numbers, among other things, Achilles does not catch up with the tortoise, and the earth cannot turn on its axis. But, as Galileo remarked, it does. It would seem to follow that something is wrong with our attempts to compass the infinite. (22) E. T. BELL

The map is not the thing mapped. When the map is identified with the thing mapped we have one of the vast melting pots of numerology. (604) E. T. BELL

The theory of numbers is the last great uncivilized continent of mathematics. It is split up into innumerable countries, fertile enough in themselves, but all more or less indifferent to one another's welfare and without a vestige of a central, intelligent government. If any young Alexander is weeping for a new world to conquer, it lies before him. (23) E. T. BELL

The present work—namely, the building of a *non-aristotelian system,* and an introduction to a theory of sanity and general semantics—depends, fundamentally, for its success on the recognition of mathematics as a language similar in structure to the world in which we live.

The maze of often unconnected knowledge we have gathered in the fields with which this part is dealing is so tremendous that it would require several volumes to cover the field even partially. Under such conditions, it is impossible to deal with the subject in any other way than by very careful selection, and so I shall, therefore, say only as much as is necessary for my present semantic purpose.

It is a common experience of our race that with a happy generalization many unconnected parts of our knowledge become connected; many 'mysteries' of science become simply a linguistic issue, and then the mysteries vanish. New generalizations introduce new *attitudes* (evaluation) which, as usual, seriously simplify the problems for a new generation. In the present work, we are treating problems from the point of view of such a generalization, of wide application; namely, *structure,* which is forced upon us by the denial of the 'is' of identity.; so that structure becomes the only link between the objective and verbal levels. The next consequence is that structure alone is the only possible content of knowledge.

Investigating structure, we have found that structure can be defined in terms of relations; and the latter, for special purposes, in terms of multi-dimensional order. Obviously, to investigate structure, we must look for relations, and so for multi-dimensional order. The full application of the above principles becomes our guide for future enquiry.

In the recent past, we have become accustomed to such arguments as, for instance, that the theory of Einstein has to be accepted on 'epistemological' grounds. Naturally, the scientist or the layman who has heard the last term, but never bothered to ascertain that it means 'according to the structure of human knowledge', would recognize no necessity to accept something which violates all his habitual *s.r*, for reasons about which he does not know or care. But if we say that the Einstein theory has to be accepted, for the 'time' being, at least, as an irreversible *structural linguistic progress,* this statement carries for many quite a different verbal and semantic implication, and one worth considering.

Mathematics has, of late, become so extremely elaborate and complex that it takes practically a lifetime to specialize in even one of its many fields. Here and there notions of extreme creative generality appear, which help us to see relations and dependence between formerly non-connected fields. For instance, the arithmetization of mathematics, or the theory of groups, or the theory of aggregates, has each become such a supreme generalization. At present, there is a general tendency among all of us, scientists included, to confuse orders of abstractions. This results in a psycho-logical semantic blockage and in the impossibility of seeing broader issues clearly.

Some of the structural issues are still but little understood, and, in writing this chapter, I lay myself open to a reproach from the layman that I have given too much attention to mathematics, and from the professional mathematician that I have given too little. My reply is that what is said here is necessary for rounding up the semantic foundations of the system, and that I explain only enough to carry the main points of structure and as semantic suggestions for further semantic researches.

I have found that among some physicists and some mathematicians the thesis that mathematics is the only language which, in 1933, is similar in structure to the world, is not always acceptable. As to the second thesis, the similarity of its structure to our nervous system, some even seem to feel that this statement borders on the sacrilegious! These objectors, apparently, believe that I ascribe more to mathematics than is just. Some physicists point out to me the non-satisfactory development of mathematics, and they seem to confuse the inadequacy of a given mathematical theory with the general *m.o* structure of mathematics. Thus, if some physical experimental investigation is conducted—for

instance, on high pressure—and the older theories predict a behaviour exemplified by the curve (A), while the experimental new data show that the actual curve is (B), such a result would show unmistakably that the first theory is not structurally correct. But, in itself, this result does not affect the correctness of a statement about the general structure of mathematics which can account for *both curves.*

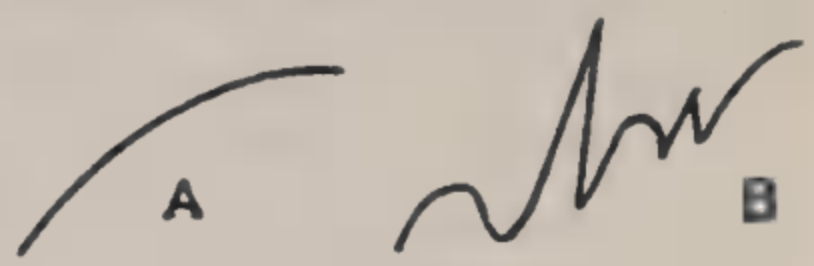

FIG. 1.

Until very lately, we had a very genuine problem in physics with the quantum phenomena which seem to proceed by discrete steps, while our mathematics is fundamentally based on assumptions of continuity. Here we had seemingly a serious structural discrepancy, which, however, has been satisfactorily overcome by the wave theory of the newer quantum mechanics, explained in Part X, where the discontinuities are accounted for, in spite of the use of differential equations and, therefore, of continuous mathematics.

But, if we start with fundamental assumptions of continuity, we always can account for discontinuities by introducing wave theories or some similar devices. Therefore, it is impossible, in our case, to argue from the wave theory (for instance) to the structure of mathematics, or vice versa, without a fundamental and *independent* general structural analysis, which alone can elucidate the problem at hand.

Mathematicians may object on the ground that the new revision of the foundation of mathematics, originated by Brouwer and Weyl, challenges the 'existence' of irrational numbers . , and, therefore, destroys the very foundations of continuity and the legitimacy of existing mathematics.

In answer to such a criticism, we should notice, first, that the current 'continuity' is of two kinds. One is of a higher grade, and is usually called by this name; the other continuity is of a lower grade and is usually called 'compactness'. The new revision challenges the higher continuity, but does not affect compactness, which, as a result, will, perhaps, have to suffice in the future for all mathematics, since compactness is sufficient to meet all psycho-logical requirements, once the problems of 'infinity' are properly understood.

A structural independent analysis of mathematics, treated as a language and a form of human behaviour, establishes the similarity of this language to the undeniable structural characteristics of this world and of the human nervous system. These few and simple structural foundations are arrived at by inspection of known data and may be considered as well established.

The existing definitions of mathematics are not entirely satisfactory. They are either too broad, or too narrow, or do not emphasize enough the main characteristics of mathematics. A semantic definition of mathematics should be broad enough to cover all existing branches of mathematics; should be narrow enough to exclude linguistic disciplines which are not considered mathematical by the best judgement of specialists, and should also be *flexible* enough to remain valid, no matter what the future developments of mathematics may be.

I have said that mathematics is the only language, at present, which in structure, is similar to the structure of the world *and* the nervous system. For purposes of exposition, we shall have to divide our analysis accordingly, remembering, in the meantime, that this division is, in a way, artificial and optional, as the issues overlap. In some instances, it is really difficult to decide under which division a given aspect should be analysed. The problems are very large, and for full discussion would require volumes; so we have to limit ourselves to a suggestive sketch of the most important aspects necessary for the present investigation.

From the point of view of general semantics, mathematics, having symbols and propositions, must be considered as a language. From the psychophysiological point of view, it must be treated as an activity of the human nervous system and as a form of the behaviour of the organisms called humans.

All languages are composed of two kinds of words: (1) Of *names* for the somethings on the un-speakable level, be they external objects., or *internal feelings,* which admittedly are *not* words, and (2) of *relational terms,* which express the actual, or desired, or any other relations between the un-speakable entities of the objective level.

When a 'quality' is treated physiologically as a reaction of an organism to a stimulus, it also becomes a relation. It should be noticed that often some words can be, and actually are, used in both senses; but, in a given context, we can always, by further analysis, separate the words used into these two categories. Numbers are not exceptions; we can use the labels 'one', 'two'., as numbers (of which the character will be explained presently) but also as names for anything we want, as, for instance, Second or Third Avenue, or John Smith I or John Smith II. When we use numbers as names, or labels for anything, we call them numerals; and this is *not* a mathematical use of 'one', 'two'., as these names do not follow mathematical rules. Thus, Second Avenue and Third Avenue cannot be added together, and do not give us Fifth Avenue in any sense whatever.

Names alone do not produce propositions and so, by themselves, say nothing. Before we can have a proposition and, therefore, meanings,

the names must be related by some relation-word, which, however, may be explicit or implied by the context, the situation, by established habits of speech, . The division of words into the above two classes may seem arbitrary, or to introduce an unnecessary complication through its simplicity; yet, if we take modern knowledge into account, we cannot follow the grammatical divisions of a primitive-made language, and such a division as I have suggested above seems structurally correct in 1933.

Traditionally, mathematics was divided into two branches: one was called arithmetic, dealing with numbers; the other was called geometry, and dealt with such entities as 'line', 'surface', 'volume', . Once Descartes, lying in bed ill, watched the branches of a tree swaying under the influence of a breeze. It occurred to him that the varying distances of the branches from the horizontal and vertical window frames could be expressed by numbers representing measurements of the distances. An epoch-making step was taken: geometrical relations were expressed by numerical relations; it meant the beginning of analytical geometry and the unification and arithmetization of mathematics.

Further investigation by the pioneers Frege, Peano, Whitehead, Russell, Keyser, and others has revealed that 'number' can be expressed in 'logical terms'—a quite important discovery, provided we have a valid 'logic' and structurally correct *non-el* terms.

Traditionally, too, since Aristotle, and, in the opinion of the majority, even today, mathematics is considered as uniquely connected with quantity and measurement. Such a view is only partial, because there are many most important and fundamental branches of mathematics which have nothing to do with quantity or measurement—as, for instance, the theory of groups, analysis situs, projective geometry, the theory of numbers, the algebra of 'logic', .

Sometimes mathematics is spoken about as the science of relations, but obviously such a definition is too broad. If the only content of knowledge is structural, then relations, obvious, or to be discovered, are the foundation of all knowledge and of *all* language, as stated in the division of words given above. Such a definition as suggested would make mathematics co-extensive with *all* language, and this, obviously, is not the case.

Before offering a semantic definition of mathematics, I introduce a synoptical table taken from Professor Shaw's *The Philosophy of Mathematics,* which he calls only suggestive and 'doubtless incomplete in many ways'. I use this table because it gives a modern list of the most important mathematical terms and disciplines necessary for the purpose of this work, indicating, also, in a way, their evolution and structural interrelations.

Central Principles of Mathematics*

MATHEMATICS ≡ QUALITATIVE STUDY OF STRUCTURE

MOTRICITIES | PATTERNS

Kinematic Mathematics | Static Mathematics

		WORLDS	MORPHOLOGY	INVARIANCY	FUNCTIONALITY	IDEALITY
Ordered	Rhythm Patterns—Arithmetic	Integers Rationals Irrationals Ensembles ATOMICITY — Numbers (Objects)	Integers Ensemble theory Literal arithmetic Complexes Point geometries Functional spaces	Congruences Arithmetic invariants Modular geometry	Arithmetic functions Algebraic functions Functions of real variable Infinitesimal calculus General analysis	Arithmetic ideals Galois ideals Higher number theory Picard-Vessiot theory
Chaotic	Space Patterns—Geometry	Points Lines Surfaces Varieties Higher elements CONTINUITY — Figures	Point space of two or more dimensions Line geometry Surface geometry Absolute geometry Higher elements Expansions	Geometric invariants Algebraic invariants Symmetric and alternating forms Modular systems	Real functions of *N* variables Vector fields Functions of lines Partial derivatives Differential geometry	Systems of differential equations Mathematical physics Relativity Theory
Ordered	Design Patterns—Tactic	Arrangements Configurations Constellations DESIGN — Forms	Combinatory analysis Stereochemistry Theory of symbols Poincaré's theory of differential equations	Stable systems Irreducibles	Functions of arrangements, configurations, constellations, and structures	Ideal elements of construction Mathematical chemistry
Ordered	Idea Patterns—Logistic	Concepts Relatives CONSISTENCY — Propositions	Foundations Postulational systems Calculus of classes, or of relatives	Equivalent systems Logical invariants	Classes functions of classes Functions of relatives Implications	Ideal classes or relatives Classificatory schemes
Chaotic	Mutation Motricities—Operators	Substitutions Transformations Groups MUTATION — Operators	Theory of finite or infinite groups Calculus of operations	Projective geometry Inversion geometry Differential and integral invariants Analysis situs	Geometric transformations Homomorphisms Transmutations	Automorphic functions Functional equations Calculus of variations Functional analysis Integral equations Difference equations Differential equations
Chaotic	Quality Motricities—Algebras	Negatives Imaginaries Hypernumbers QUALITY — Vids	Linear algebras Functional transformations	Invariant equations Invariants of expressions	Functions of complex and hypercomplex variables General function theory	Hyperideals
Ordered	Action Motricities—Processes	Routes Displacements Combinations ACTION — Processes	Composite actions Actional structures	Equivalents of actions Invariants of processes	Processes dependent on processes Functions of action	Ideal processes
Inferences		Transpositions Syllogisms Implications DEDUCTION — Thoughts	Syllogisms Calculus of deduction	Laws of inference Equivalent deductions Invariants	Functions of inferences	Ideal entities that satisfy inferences Scientific theories

*This table differs slightly from the one printed in *The Philosophy of Mathematics*. The corrections have been made by Professor Shaw and kindly communicated to me by letter.

A semantic definition of mathematics may run somehow as follows: Mathematics consists of limited linguistic schemes of multiordinal relations capable of exact treatment at a given date.

After I have given a semantic definition of number, it will be obvious that the above definition covers all existing disciplines considered mathematical. However, these developments are not fixed affairs. Does that definition provide for their future growth? By inserting as a fundamental part of the definition 'exact treatment at a given date', it obviously does. Whenever we discover any relations in any fields which will allow exact 'logical' treatment, such a discipline will be included in the body of linguistic schemes called mathematics, and, at present, there are no indications that these developments can ever come to an end. When 'logic' becomes an ∞-valued 'structural calculus', then mathematics and 'logic' will merge completely and become a general science of *m.o* relations and multi-dimensional order, and all sciences may become exact.

It is necessary to show that this definition is not too broad, and that it eliminates notions which are admittedly non-mathematical, without invalidating the statement that the content of all knowledge is structural, and so ultimately relational. The word 'exact' eliminates non-mathematical relations. If we enquire into the meaning of the word 'exact', we find from experience that this meaning is not constant, but that it varies with the date, and so only a statement 'exact at a given date' can have a definite meaning.

We can analyse a simple statement, 'grass is green' (the 'is' here is the 'is' of predication, not of identity), which, perhaps, represents an extreme example of a non-mathematical statement; but a similar reasoning can be applied to other examples. Sometimes we have a feeling which we express by saying, 'grass is green'. Usually, such a feeling is called a 'perception'. But is such a *process* to be dismissed so simply, by just calling it a name, 'perception'? It is easy to 'call names under provocation', as Santayana says somewhere; but does that exhaust the question?

If we analyse such a statement further, we find that it involves comparison, evaluation in certain respects with other characters of experience., and the statement thus assumes relational characteristics. These, in the meantime, are non-exact and, therefore, non-mathematical. If we carry this analysis still further, involving data taken from chemistry, physics, physiology, neurology., we involve relations which become more and more exact, and, finally, in such terms as 'wave-length', 'frequency'., we reach structural terms which allow of exact 1933 treatment. It is true that a language of 'quality' conceals relations, sometimes very effectively; but once 'quality' is taken as the reaction of a given organism

to a stimulus, the term used for that 'quality' becomes a name for a very complex relation. This procedure can be always employed, thus establishing once more the fundamental character of relations.

These last statements are of serious structural and semantic importance, being closely connected with the $\bar{A}$, fundamental, and undeniable negative premises. These results can be taught to children very simply; yet this automatically involves an entirely new and modern method of evaluation and attitude toward language, which will affect beneficially the, as yet entirely disregarded, *s.r.*

We must consider, briefly, the terms 'kind' and 'degree', as we shall need them later. Words, symbols . , serve as forms of representation and belong to a different universe—the 'universe of discourse'—since they are not the un-speakable levels we are speaking about. They belong to a world of higher abstractions and not to the world of lower abstractions given to us by the lower nerve centres.

Common experience and scientific investigations (more refined experience) show us that the world around us is made up of absolute individuals, each different and unique, although interconnected. Under such conditions it is obviously optional what language we use. The more we use the language of diverse 'kind', the sharper our definitions must be. Psycho-logically, the emphasis is on difference. Such procedure may be a tax on our ingenuity, but by it we are closer to the structural facts of life, where, in the limit, we should have to establish a 'kind' for every individual.

In using the term 'degree', we may be more vague. We proceed by similarities, but such a treatment implies a fundamental interconnection between different individuals of a special kind. It implies a definite kind of metaphysics or structural assumptions—as, for instance, a theory of evolution. As our 'knowledge' is the result of nervous abstracting, it seems, in accordance with the structure of our nervous system, to give preference to the term 'degree' first, and only when we have attained a certain order of verbal sharpness to pass to a language of 'kind', if need arises.

The study of primitive languages shows that, historically, we had a tendency for the 'kind' language, resulting in over-abundance of names and few relation-words, which makes higher analysis impossible. Science, on the other hand, has a preference for the 'degree' language, which, ultimately, leads to mathematical languages, enormous simplicity and economy of words, and so to better efficiency, more intelligence, and to the unification of science. Thus, chemistry became a branch of physics; physics, a branch of geometry; geometry merges with analysis, and

analysis merges with general semantics; and life itself becomes a physico-chemical colloidal occurrence. The language of 'degree' has very important *relational, quantitative,* and *order* implications, while that of 'kind' has, in the main, qualitative implications, often, if not always, concealing relations, instead of expressing them.

The current definition of 'number', as formulated by Frege and Russell, reads: 'The number of a class is the class of all those classes that are similar to it'.[1] This definition is not entirely satisfactory: first, because the multiordinality of the term 'class' is not stated; second, it is *A*, as it involves the ambiguous (as to the order of abstractions) term 'class'. What do we mean by the term 'class'? Do we mean an extensive array of absolute individuals, un-speakable by its very character, such as some *seen aggregate,* or do we mean the *spoken definition* or *description* of such un-speakable objective entities? The term implies, then, a fundamental confusion of orders of abstractions, to start with—the very issue which we must avoid most carefully, as positively demanded by the non-identity principle. Besides, if we explore the world with a 'class of classes'., and obtain results also of 'class of classes', such procedure throws no light on mathematics, their applications and their importance as a tool of research. Perhaps, it even increases the mysteries surrounding mathematics and conceals the relations between mathematics and human knowledge in general.

We should expect of a satisfactory definition of 'number' that it would make the semantic character of numbers clear. Somehow, through long experience, we have learned that numbers and measurement have some mysterious, sometimes an uncanny, importance. This is exemplified by mathematical predictions, which are verified later empirically. Let me recall only the discovery of the planet Neptune through mathematical investigations, based on its action upon Uranus, long before the astronomer actually verified this prediction with his telescope. Many, a great many, such examples could be given, scientific literature being full of them. Why should mathematics and measurement be so extremely important? Why should mathematical operations of a given Smith, which often seem innocent (and sometimes silly enough) give such an unusual security and such undeniably practical results?

Is it true that the majority of us are born mathematical imbeciles? Why is there this general fear of, and dislike for, mathematics? Is mathematics really so difficult and repelling, or is it the way mathematics is treated and taught by mathematicians that is at fault? If some light can be thrown on these perplexing semantic problems, perhaps we shall face a scientific revolution which might deeply affect our educational

system and may even mark the beginning of a new period in standards of evaluation, in which mathematics will take the place which it ought to have. Certainly, there must be something the matter with our epistemologies and 'psychologies' if they cannot cope with these problems.

A simple explanation is given by a new $\bar{A}$ analysis and a *semantic* definition of numbers. What follows is written, in the main, for non-mathematicians, as the word 'semantic' indicates, but it is hoped that professional mathematicians (or some, at least) may be interested in the *meanings* of the term 'number', and that they will not entirely disregard it. As semantic, the definition seems satisfactory; but, perhaps, it is not entirely satisfactory for technical purposes, and the definition would have to be slightly re-worded to satisfy the technical needs of the mathematicians. In the meantime, the gains are so important that we should not begrudge any amount of labour in order to produce finally a mathematical and, this time, $\bar{A}$ semantic definition of numbers.

As has already been mentioned, the importance of notation is paramount. Thus the Roman notation for number—I, II, III, IV, V, VI.,—was not satisfactory and could not have led to modern developments in mathematics, because it did not possess enough positional and structural characteristics. Modern mathematics began when it was made possible by the invention or discovery of positional notation. We use the symbol '1' in 1, 10, 100, 1000., in which, because of its place, it had different values. In the expression '1', the symbol means 'one unit'; in 10, the symbol '1' means ten units; in 100, the symbol '1' means one hundred units,.

To have a positional notation, we need a symbol '0', called zero, to indicate an empty column and, at least, one symbol '1'. The number of special symbols for 'number' depends on what base we accept. Thus, in a binary system, with the base 2, our 1 is represented by 1; our 2, by a unity in the second place and a zero in the first place, thus by 10; our 3, by 11; our 4, by a unity in the third place and two zeros in the first and second places; namely, 100; our 5, by 101; our 6, by 110; 7, by 111; 8, by 1000; 9, by 1001; . . . 15, by 1111; 16, by 10000,. In a binary system, we needed only the two symbols, 1 and 0. For a system with the base 3, we would need three symbols, 1, 2, 0: our 1 would be represented by 1; our 2, by 2; 3, by 10; 4, by 11; 5, by 12; 6, by 20,. In our decimal system, obviously, we need 10 symbols, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0.

For more details on notation, the interested reader is referred to the fascinating and elementary book of Professor Danzig, *Number the Language of Science*. Here we only emphasize what is necessary for our purpose. Every system has its advantages and drawbacks. Thus, in

the binary system, still used by some savage tribes, of which we retain traces when we speak of couples, or pairs, or braces, we get an enormous simplicity in operations by using only two symbols, 1 and 0. It should be remembered that, in every system, the tables of addition and multiplication must be memorized. In the binary system, these tables are reduced to $1 + 1 = 10$ and $1 \times 1 = 1$, while in our decimal system, each table has 100 entries. But what we gain in simplicity by a low base-number is offset very seriously by the cumbersomeness of the notation. As Danzig tells us, our number 4096 is represented in a binary system by 1,000,000,000. That we adopted the decimal system is probably a physiological accident, because we have ten fingers. The savage, with his binary system, did not reach even the finger stage; he is still in the fist stage.

For practical purposes, it is simpler to have a base which has many divisors, as, for instance, 12. We still use this duodecimal system when we divide a foot into twelve inches, or a shilling into twelve pennies, or count by dozens or gross. It seems that mathematicians would probably select a prime number for a base, but the gain would be so slight and the difficulty of offsetting a physiological habit so tremendous, that this will probably never happen.[2]

From what was said already, it is, perhaps, clear that mathematics requires a positional notation in which we must have a symbol for '1' and zero, at least. For these and other reasons, the two numbers 1 and 0 are somehow unusually important. Even in our decimal system we generate numbers by adding 1 to its predecessor. Thus $1 + 1 = 2$, $2 + 1 = 3$., and we must enquire into the semantic character of these numbers.

The notions of matching, comparing, measuring, quantity, equality., are all interwoven and, by necessity, involve a circularity in definitions and implications if the analysis is carried far enough. The interested reader may be referred to the chapter on equality in Whitehead's *The Principle of Relativity* to learn more on this subject.

In the evolution of mathematics, we find that the notions of 'greater', 'equal', and 'less' precede the notion of numbers. Comparison is the simplest form of evaluation; the first being a search for relations; the second, a discovery of exact relations. This process of search for relations and structure is inherent and natural in man, and has led not only to the discovery of numbers, but also has shaped their two aspects; namely, the cardinal and the ordinal aspects. For instance, to ascertain whether the number of persons in a hall is equal to, greater than, or less than the number of seats, it is enough to ascertain if all seats are

occupied and there are no empty seats and no persons standing; then we would say that the number of persons is equal to the number of seats, and a *symmetrical relation* of equality would be established. If all seats were occupied and there were some persons standing in the hall, or if we found that no one was standing, yet not all seats were occupied, we would establish the *asymmetrical relation* of greater or less.

In the above processes, we were using an important principle; namely, that of *one-to-one* correspondence. In our search for relations, we assigned to each seat one person, and reached our conclusions without any counting. This process, based on the *one-to-one* correspondence, establishes what is called the cardinal number. It gives us specific relational data about this world; yet it is not enough for counting and for mathematics. To produce the latter, we must, first of all, establish a definite system of symbolism, based on a definite relation for generating numbers; for instance, $1+1=2$, $2+1=3$., which establishes a definite *order*. Without this ordinal notion, neither counting nor mathematics would be possible; and, as we have already seen, order can be used for defining relations, as the notions of relation and order are interdependent. Order, also, involves asymmetrical relations.

If we consider the two most important numbers, 0 and 1, we find that in the accepted symbolism, if $a=b$, $a-b=0$; and if $a=b$, $a/b=b/b=1$; so that both fundamental numbers express, or can be interpreted as expressing, a *symmetrical relation* of equality.

If we consider any other number—and this applies to all kinds of numbers, not only to natural numbers—we find that any number is not altered by dividing it by one, thus, $2/1=2$, $3/1=3$., in general, $N/1=N$; establishing the *asymmetrical* relation, *unique* and *specific* in a given case that N is N times more than one.

If we consider, further, that $2/1=2$, $3/1=3$, *and so on,* are all *different, specific,* and *unique,* we come to an obvious and $\bar{A}$ semantic definition of number in terms of relations, in which 0 and 1 represent *unique* and *specific symmetrical* relations and all other numbers also *unique* and *specific asymmetrical* relations. Thus, if we have a result '5', we *can always say* that the number 5 is five times as many as one. Similarly, if we introduce apples. Five apples are five times as many as one apple. Thus, a number in any form, 'pure' or 'applied', can always be represented as a relation, *unique* and *specific* in a given case; and this is the foundation of the exactness of dealing with numbers. For instance, to say that *a* is greater than *b* also establishes an asymmetrical relation, but it is *not unique* and *specific;* but when we say that *a* is five

times greater than *b*, this relation is *asymmetrical, exact, unique,* and *specific.*

The above simple remarks are not entirely orthodox. That $5/1 = 5$ is very orthodox, indeed; but that numbers, in general, represent indefinitely many *exact, specific,* and *unique,* and, in the main, *asymmetrical* relations is a structural notion which necessitates the revision of the foundations of mathematics and their rebuilding on the basis of new semantics and a future structural calculus. When we say 'indefinitely many', this means, from the reflex point of view, 'indefinitely flexible', or 'fully conditional' in the semantic field, and, therefore, a prototype of human semantic reactions (see Part VI). The scope of the present work precludes the analysis of the notion of the lately disputed 'irrational'; but we must state that this revision requires new psycho-logical and structural considerations of fundamental 'logical' postulates and of the problems of 'infinity'. If, by an arbitrary process, we *postulate* the existence of a 'number' which alters all the while, then, according to the definition given here, such expressions should be considered as functions, perhaps, but not as a number, because they do not give us *unique* and *specific relations.*

These few remarks, although suggestive to the mathematician, do not, in any way, exhaust the question, which can only be properly presented in technical literature in a postulational form.

It seems that mathematicians, no matter how important the work which they have produced, have never gone so far as to appreciate fully that they are willy-nilly producing an ideal human relational language of structure similar to that of the world *and* to that of the human nervous system. This they cannot help, in spite of some vehement denials, and their work should also be treated from the semantic point of view.

Similarly with measurement. From a functional or actional and semantic point of view, measurement represents nothing else but a search for *empirical structure* by means of extensional, ordered, symmetrical, and asymmetrical relations. Thus, when we say that a given length measures five feet, we have reached this conclusion by *selecting* a unit called 'foot', an *arbitrary and un-speakable* affair, then laying it end to end five times in a definite *extensional order* and so have established the asymmetrical, and, in each case, *unique* and *specific* relation, that the given entity represents, in this case, five times as many as the arbitrarily selected unit.

Objection may be raised that the formal working out of a definition of numbers in terms of relations, instead of classes, would be very

laborious, and would also involve a revision of the foundations of mathematics. This can hardly be denied; but, in the discussions of the foundations, the confusion of orders of abstractions is still very marked, thereby resulting in the manufacture of *artificial semantic difficulties.* Moreover, the benefits of such a definition, in eliminating the mysteries about mathematics, are so important that they by far outweigh the difficulties.

As the only possible content of knowledge is structural, as given in terms of relations and multiordinal and multi-dimensional order; numbers, which establish an endless array of exact, specific, and, in each case, unique, relations are obviously the most important tools for exploring the *structure* of the world, since structure can always be analysed in terms of *relations.* In this way, all mysteries about the importance of mathematics and measurement vanish. The above understanding will give the student of mathematics an entirely different and a very natural feeling for his subject. As his only possible aim is the study of structure of the world, or of whatever else, he must naturally use a *relational tool* to explore this complex of relations called 'structure'. A most spectacular illustration of this is given in the internal theory of surfaces, the tensor calculus., described in Part VIII.

In all measurements, we select a unit of a necessary kind, for a given case, and then we find a *unique* and *specific* relation as expressed by a number, between the given something and the selected unit. By relating different happenings and processes to the same unit-process, we find, again, *unique* and *specific* interrelations, in a given case, between these events, and so gather structural (and most important, because uniquely possible) wisdom, called 'knowledge', 'science',.

If we treat numbers as relations, then fractions and all operations become relations of relations, and so relations of higher order, into the analysis of which we cannot enter here, as these are, of necessity, technical.

It should be firmly grasped, however, that some fundamental human relations to this world have not been changed. The primitive may have believed that words *were* things (identification) and so have established what is called the 'magic of words' (and, in fact, the majority of us still have our *s.r* regulated by some such unconscious identifications); but, in spite of this, the primitive or 'civilized' man's words *are not,* and never could be, the things spoken about, no matter what semantic disturbances we might have accompanying their use, or what delusions or illusions we may cherish in respect to them.

At present, of all branches of mathematics the theory of numbers is probably the most difficult, obscure., and seemingly with the fewest applications. With a new $\bar{A}$ definition of numbers in terms of relations, this theory may become a relational study of very high order, which, perhaps, will some day become the foundation for epistemology and the key for the solution of all the problems of science and life. In the fields of cosmology many, if not the majority, of the problems, by necessity, cannot be considered as directly experimental, and so the solution must be epistemological.

At present, in our speculations, we are carried away by words, disregarding the simple fact that speaking about the 'radius of the universe', for instance, has *no meaning,* as it cannot possibly be observed. Perhaps, some day, we shall discover that such conversations are the result of our old stumbling block, identification, which leads to our being carried away by the sounds of words applicable to terrestrial conditions but meaningless in the very small, as discovered lately in the newer quantum mechanics, and, in the very large, as applied to the cosmos. An important illustration of the retardation of scientific progress, blocked by the confusion of orders of abstractions, is shown in the fact that the newer quantum mechanics were slow in coming, and though astronomers probably know about it, yet they still fail to grasp that expressions such as the 'radius of the universe', the 'running down of the universe'., are meaningless outside of psychopathology.

In this connection, we should notice an extremely interesting and important semantic characteristic; namely, that the term 'relation' is not only multiordinal but also *non-el,* as it applies to 'senses' and 'mind'. Relations are usually found empirically; so in a language of relations we have a language of similar structure to the world and a unique means for predictability and rationality.

Let me again emphasize that, from time immemorial, things have *not* been words; the only content of knowledge has been structural; mathematics has dealt, in the main, with numbers; no matter whether we have understood the character of numbers or not, numbers have expressed relations and so have given us structural data willy-nilly,. This explains why mathematics and numbers have, since time immemorial, been a favorite field, not only for speculations, but also why, in history, we find so many religious semantic disturbances connected with numbers. Mankind has somehow felt instinctively that in numbers we have a potentially endless array of *unique* and *specific exact relations,* which ultimately give us structure, the last being the only possible content of knowledge, because words are *not* things.

As relations, generally, are empirically present, and as man and his 'knowledge' is as 'natural' as rocks, flowers, and donkeys, we should not be surprised to find that the unique language of exact., relations called mathematics is, by necessity, the natural language of man and *similar in structure* to the world *and* our nervous system.

As has already been stated, it is incorrect to argue from the structure of mathematical theories to the structure of the world, and so try to establish the similarity of structure; but that such enquiry must be *independent* and start with quite ordinary structural experiences, and only at a later stage proceed to more advanced knowledge as given by science. Because this analysis must be independent, it can also be made very simple and elementary. All exact sciences give us a wealth of experimental data to establish the first thesis on similarity of structure; and it is unnecessary to repeat it here. I will restrict myself only to a minimum of quite obvious facts, reserving the second thesis—about the similarity of structure with our nervous system—for the next chapter.

If we analyse the silent objective level by objective means available in 1933, say a microscope, we shall find that whatever we can see, handle., represents an *absolute individual,* and *different from anything else in this world.* We discover, thus, an important *structural* fact of the external world; namely, that in it, everything we can see, touch., that is to say, all lower order abstractions represent absolute individuals, different from everything else.

On the verbal level, under such empirical conditions, we should then have a language of *similar structure;* namely, one giving us an indefinite number of *proper names, each different.* We find such a language *uniquely* in numbers, each number 1, 2, 3., being a unique, sharply distinguishable, *proper name* for a relation, and, if we wish, for anything else also.

Without some higher abstractions we cannot be human at all. No science could exist with absolute individuals and no relations; so we pass to higher abstractions and build a language of say x_i, $(i = 1, 2, 3, \ldots n)$, where the x shows, let us say, that we deal with a variable x with many values, and the number we assign to i indicates the individuality under consideration. From the structural point of view, such a vocabulary is similar to the world around us; it accounts for the individuality of the external objects, it also is similar to the structure of our nervous system, because it allows generalizations or higher order abstractions, emphasizes the abstracting nervous characteristics,. The subscript emphasizes the differences; the letter x implies the similarities.

In daily language a similar device is extremely useful and has very far-reaching psycho-logical semantic effects. Thus, if we say 'pencil_1', 'pencil_2', . . . 'pencil_n', we have indicated structurally two main characteristics: (1) the absolute individuality of the object, by adding the indefinitely individualizing subscript 1, 2, . . . n; and (2) we have also complied with the nervous higher order abstracting characteristics, which establish similarity in diversity of different 'pencils'. From the point of view of *relations,* these are usually found empirically; besides, they may be invariant, no matter how changing the world may be.

In general terms, the structure of the external world is such that we deal always on the objective levels with absolute individuals, with absolute differences. The structure of the human nervous system is such that it abstracts, or generalizes, or integrates., in higher orders, and so finds similarities, discovering often invariant (sometimes relatively invariant) relations. To have 'similar structure', a language should comply with both structural exigencies, and this characteristic is found in the mathematical notation of x_i, which can be enlarged to the daily language as 'Smith_i', 'Fido_i'., where $i = 1, 2, 3, \ldots n$.

Further objective enquiry shows that the world and ourselves are made up of *processes,* thus, 'Smith_{1900}' is quite a different person from 'Smith_{1933}'. To be convinced, it is enough to look over old photographs of ourselves, the above remark being structurally entirely general. A language of 'similar structure' should cover these facts. We find such a language in the vocabulary of 'function', 'propositional function', as already explained, involving also four-dimensional considerations.

As words are not objects—and this expresses a structural fact—we see that the 'is' of identity is unconditionally false, and should be entirely abolished as such. Let us be simple about it. This last semantic requirement is genuinely difficult to carry out, because the general *el* structure of our language is such as to facilitate identification. It is admitted that in some fields some persons identify only a little; but even they usually identify a great deal when they pass to other fields. Even science is not free from identification, and this fact introduces great and artificial semantic difficulties, which simply vanish when we stop identification or the confusion of orders of abstractions. Thus, for instance, the semantic difficulties in the foundations of mathematics, the problems of 'infinity', the 'irrational'., the difficulties of Einstein's theory, the difficulties of the newer quantum theory, the arguments about the 'radius of the universe', 'infinite velocities', the difficulties in the present theory., ., are due, in the main, to semantic blockages or commitment to the structure of the old language—we may call it 'habit'—*which says structurally very little,*

and which I disclose as a semantic disturbance of *evaluation* by showing the *physiological mechanism in terms of order.*

If we abolish the 'is' of identity, then we are left only with a functional, actional., language elaborated in the mathematical language of function. Under such conditions, a *descriptive* language of ordered happenings on the objective level takes the form of 'if so and so happens, then so and so happens', or, briefly, 'if so, then so'; which is the prototype of 'logical' and mathematical processes and languages. We see that such a language is again similar in structure to the external world descriptively; yet it is similar to the 'logical' nervous processes, and so allows us, because of this similarity of structure, predictability and so rationality.

In the traditional systems, we did not recognize the complete semantic interdependence of differences and similarities, the empirical world exhibiting differences, the nervous system manufacturing primarily similarities, and our 'knowledge', if worth anything at all, being the *joint product* of both. Was it not Sylvester who said that 'in mathematics we look for similarities in differences and differences in similarities'? This statement applies to our whole abstracting process.

The empirical world is such in structure (by inspection) that in it we can add, subtract, multiply, and divide. In mathematics, we find a language of similar structure. Obviously, in the physical world these actions or operations alter the relations, which are expressed as altered unique and specific relations, by the language of mathematics. Further, as the world is full of different shapes, forms, curves., we do not only find in mathematics special languages dealing with these subjects, but we find in analytical geometry unifying linguistic means for translation of one language into another. Thus any 'quality' can be formulated in terms of relations which may take the 'quantitative' character which, at present, in all cases, can be also translated into geometrical terms and methods, giving structures to be *visualized.*

It is interesting, yet not entirely unexpected, that the activities of the higher nervous centres, the conditional reflexes of higher order, the semantic reactions, time-binding included, should follow the exponential rules, as shown in my *Manhood of Humanity.*

In our experience, we find that some issues are additive—as, for instance, if one guest is added to a dinner party, we will have to add plates and a chair. Such facts are covered by additive methods and the language called 'linear' (see Part VIII). In many instances—and these are, perhaps, the most important and are strictly connected with submicroscopic processes—the issues are not additive, one atom of oxygen

'plus' two atoms of hydrogen, under proper conditions, will produce water, of which the characteristics are *not* the sum of the characteristics of oxygen and hydrogen 'added' together, but entirely *new* characteristics *emerge*. These may some day be taken care of by non-linear equations, when our knowledge has advanced considerably. These problems are unusually important and vital, because with our present low development and the lack of structural researches, we still keep an additive *A* language, which is, perhaps, able to deal with additive, simple, immediate, and comparatively unimportant issues, but is entirely unfit structurally to deal with principles which underlie the most fundamental problems of life. Similarly, in physics, only since Einstein have we begun to see that the primitive, simplest, and easiest to solve linear equations are not structurally adequate.

One of the most marked structural characteristics of the empirical world is 'change', 'motion', 'waves', and similar dynamic manifestations. Obviously, a language of similar structure must have means to deal with such relations. In this respect, mathematics is unique, because, in the differential and integral calculus, the four-dimensional geometries and similar disciplines, with all their developments, we find such a perfect language to be explained more in detail in the chapters which follow.

It will be profitable for our purpose to discuss, in the next chapter, some of the mathematical structural characteristics in connection with their similarity to the human nervous system; but here I will add only that, for our purposes, at this particular point, we must specially emphasize *arithmetical* language, which means numbers and arithmetical operations, the theory of function, the differential and integral calculus (language) and different geometries in their two aspects, 'pure' and applied. Indeed, Riemann tells us bluntly that the *science* of physics originated only with the introduction of differential equations, a statement which is quite justified, but to which I would add, that physics is becoming scientific since we began to eliminate from physics semantic disturbances; namely, identification and elementalism. This movement was originated, in fact, although not stated in an explicit form, by the Einstein theory and the new quantum theories, the psycho-logical trend of which is formulated in a *general semantic theory* in the present work.

It is reasonable to consider that metric geometry, and, in particular, the *E*-system, was derived from touch, and, perhaps, the 'kinesthetic sense', and projective geometry from sight.

Although the issues presented here appear extremely simple, and sometimes even commonplace, yet the actual working out of the verbal

schemes is quite elaborate and ingenious, and impossible to analyse here more fully; so that only one example can be given.

The solution of mathematical equations is perhaps to be considered as the central problem of mathematics. The word 'equation' is derived from Latin *aequare,* to equalize, and is a statement of the symmetrical relation of equality expressed as $a = b$ or $a - b = 0$ or $a/b = 1$. An equation expresses the relation between quantities, some of which are known, some unknown and to be found. By the solution of an equation, we mean the finding of values for the unknowns which will satisfy the equation.

Linear equations of the type $ax = b$ necessitated the introduction of fractions. Linear equations with several variables led to the theory of determinants and matrices . , which underwent, later, a tremendous independent development; yet they originated in the attempt to simplify the solution of these equations.

Quadratic equations of the type $x^2 + ax + b = 0$ can be reduced to the form $z^2 = A$, the solution of which depends on the extraction of a square root. Here, serious difficulties arose, and seemingly necessitated the introduction of 'irrational' numbers and ordinary complex numbers, involving the square root of minus one ($\sqrt{-1} = i$), a notion which revolutionized mathematics.

Cubic equations of the form $x^3 + ax^2 + bx + c = 0$ necessitated the extraction of cube roots, in addition to the problems connected with the solution of quadratic equations, and involved more difficulties, which have been analysed in an extensive literature.

Biquadratic equations of the type $x^4 + ax^3 + bx^2 + cx + d = 0$ involve the problems of quadratic and cubic equations. When we consider equations of a degree higher than the fourth, we find that we cannot solve them by former methods; and mathematicians have had to invent theories of substitutions, groups, different special functions and similar devices. The solution of differential equations introduced further difficulties, allied with the theory of function.

The linear transformations of algebraic polynomials with two or more variables in connection with the theory of determinants, symmetrical functions, differential operations . , necessitated the development of an extensive theory of algebraic forms which, at present, is far from being complete.

In the above analysis, I have refrained from giving details, most of which would be of no value to the layman, and unnecessary for the mathematician; but it must be emphasized that the theory of function and the theory of groups, with their very extensive developments,

involving the theory of invariance, and, in a way, the theory of numbers, rapidly became a unifying foundation upon which practically the whole of mathematics is being rebuilt. Many branches of mathematics have become, of late, nothing more than a theory of invariance of special groups.

As to practical applications, there is no possibility to list them, and the number increases steadily. But, without the theory of analytic function, for instance, we could not study the flow of electricity, or heat, or deal with two-dimensional gravitational, electrostatic, or magnetic attractions. The complex number involving the square root of minus one was necessary for the development of wireless and telegraphy; the kinetic theory of gases and the building of automobile engines require geometries of n dimensions; rectangular and triangular membranes are connected with questions discussed in the theory of numbers; the theory of groups has direct application in crystallography; the theory of invariants underlies the theory of Einstein, the theory of matrices and operators has revolutionized the quantum theory; and there are other applications in an endless array.[3]

In Part VIII, different aspects of mathematics are analysed, but the interested reader can be referred also to the above-mentioned book of Professor Shaw for an excellent elementary, yet structural, view of the progress of mathematics.